DiVA
http://www.diva-portal.org

Postprint

This is the accepted version of a paper presented at *ASIS&T 2023 Annual Meeting*.

Citation for the original published paper:

Eriksson-Backa, K., Enwald, H., Hirvonen, N., Huvila, I. (2023)
Involving Older Users in Design of Patient Accessible Electronic Health Records: Exploring the Potential of Open-Ended Questions
In: (pp. 576-580). Silver Spring, MD
Proceedings of the Association for Information Science and Technology
https://doi.org/10.1002/pra2.818

N.B. When citing this work, cite the original published paper.

Permanent link to this version:
http://urn.kb.se/resolve?urn=urn:nbn:se:uu:diva-519572

Involving Older Users in Design of Patient Accessible Electronic Health Records: Exploring the Potential of Open-ended Questions

AUTHORS SECTION

ABSTRACT

Despite older adults being one of the most important groups of users of eHealth services, including patient accessible electronic health records, these are claimed not to be designed to suit older users, and to not necessarily fulfill their information needs. User-centered design methods often utilized when developing eHealth services have also been claimed not to work well for older adults. This paper suggests an alternative way of collecting ideas and feedback on eHealth from older adults, by including open-ended questions in the data collection of larger studies on eHealth information behavior. When data are collected by methods like focus group interviews and surveys, older adults can be requested to envision an eHealth service that suits their needs. Results from studies in Finland are presented, and show that current eHealth services do not fulfill all information needs of older adults, as answers to the question in both focus group interviews and surveys contain practical suggestions for additional and more comprehensible information, easier ways of searching for needed information, and alternative channels for communication. Open-ended questions can be a cost-effective and timesaving way to help in the design and development of patient accessible electronic health records and other types of eHealth services.

KEYWORDS

eHealth, focus groups, older adults, service development, surveys

INTRODUCTION

Older adults are generally thought to benefit from eHealth solutions such as patient portals (e.g. Wildenbos et al., 2018), but older age groups might not use eHealth applications if these are not perceived to fulfill the users' needs or the users experience difficulties in using them (Vergouw, 2020). In library and information science, the aim of research is generally to come up with solutions to fulfill information needs of users or potential users, be it of existing or planned services, systems or products within these. This has been at the core of the field at least since the early 1980s, when Tom Wilson published his much-cited work on information needs and user studies (Wilson, 1981). Later, Mark Hepworth (Hepworth, 1997) underlined the need to understand the consumers in the context of design and development of electronic information products and services. The focus of research has often been to suggest how knowledge of users' information behavior can be useful for designing information systems in various contexts ranging from archeology to health. Yet, information behavior researchers have often failed to reach out to systems developers, for various reasons (Huvila et al. 2022). Rytkönen and colleagues (Rytkönen, Kinnunen & Martikainen, 2022), however, found that Finnish software developers working with information systems within social and healthcare were positive towards end-user input, as they considered feedback useful, but cooperation with users was restrained by lack of established methods for cooperation and necessary resources. In addition, the end-users the software developers considered important to work with were mainly health professionals or people in decision-making positions, not laypersons. Comprehensive user-centered design of new systems is, in addition, usually time-consuming, and there is room for other and faster methods to examine users' needs and preferences. This paper suggests involving older laypersons in innovating eHealth services that suit their needs, and presents findings from a larger study on eHealth information behavior of older adults in Finland.

MOTIVATION

The focus of this paper is on patient accessible electronic health records (PAEHR). While electronic health records are records and systems maintained by health professionals and official agencies (Comandé, Nocco & Peigné 2015), a PAEHR is a service that is provided by an online system and that grants patients continuous access to their medical health data from either electronic health records or other systems related to health (Essén et al 2018). The value of PAEHRs lies mainly in the possibility to check test results, and for renewal of prescriptions and secure messaging with healthcare personnel (Luo, Dozier, & Ikenberg, 2021). Reviewing one's medical record and checking appointments are also perceived to be beneficial (Wildenbos et al., 2018). Frustration with technical issues

86th Annual Meeting of the Association for Information Science & Technology | Oct. 29 – 31, 2023 |London, United Kingdom. *Author(s) retain copyright, but ASIS&T receives an exclusive publication license.*

related to access, temporal unavailability of information, or limitations in the available contents, on the other hand, possibly limits use. Limitations in the available content can be an issue especially for those who want to know more about particular aspects of their health or treatment, for example, test results (Shah et al.,2015). Timeliness can be a problem when the content does not become available fast enough. For example, test results are often requested to be available quickly. Usability issues, such as difficulties with the login process, form additional barriers to use (Shah et al,.2015; Wildenbos et al., 2018).

Older adults are perhaps the most important group of PAEHR users. However, it is claimed that health technologies fail to be designed in a way that meets the expectations of those in older age groups (Wildenbos et al., 2018). Stojmenova et al. (2012) feared that if users are not involved in the design of eHealth services, this might influence how well they accept and adopt these services. In order to involve end-users, user-centered design has been considered crucial for eHealth design, for example, to increase user satisfaction and adoption (van Velsen, Ludden, & Grünloh, 2022). Gould and Lewis (1985) describe user-centered design as a situation where designers study users and what they need to do with technology. For this purpose, users need to work with existing prototypes, and the technology should be tested and possibly redesigned several times during the process. Over the years, new approaches have been introduced to address this issue, including activity-centered, community-driven and economic design approaches, and citizen science and participatory action. Researchers who have worked with user-centered design have also expressed concerns about, for example, sampling bias, limited and biased end-user input, overreliance on end-user input, that end users are only a subset of those who should be heard, and that ethical, societal and political aspects are missed out (van Velsen, Ludden, & Grünloh, 2022). In addition, this approach requires time and effort. Stojmenova et al. (2012), on their hand, claimed that conventional user-centered design did not work well with older adults, due to various factors. It is hence motivated to try out other methods.

AIM

Drawing on earlier research showing the importance of involving end-users in development of eHealth applications, and existing problems in doing so, this paper explores how (potential) users can be involved in development or improvement of PAEHRs in a way that is simpler than the user-centered design approach. We suggest using a method based on asking users open-ended questions on what their ideal eHealth service or application would look like, and which features it would have, as a part of focus group interviews and surveys examining eHealth information behavior more extensively.

METHODS

As part of a larger Finnish research project aiming at connecting health information behavior research and development of health technologies for older adults, two sets of data were collected separately. The methods were in part overlapping with those used in user-centered design; focus group interviews and surveys (Stojmenova, Imperl, Žohar & Dinevski, 2012). Focus groups have been described as a good way to examine needs, wishes and opinions on eHealth (Kip et al., 2022). Suggestions developing during the discussions in the focus groups can be more in-depth, and more varied than those in surveys, as more people with more ideas are involved in the discussion, whereas the answers in surveys are generally the output of one single person who has to write down the answer, that can be quite short and not much elaborated on. Questionnaires are hence not suggested to be the sole method for collecting data, but triangulation has been suggested (Kip et al., 2022). On the other hand, the number of survey answers is much larger and can give more variation. In contrast to previous mixed-method studies such as that by Papoutsi et al. (2015), where survey results were later discussed in focus groups, these particular studies were conducted separately.

Focus group interviews were conducted with a convenience sample consisting of 24 persons aged 55-73 years divided into six groups, and printed questionnaires were sent by mail to a nation-wide sample of 1,500 Finnish persons aged 55-70 years. This sample was randomly drawn from the population register system of Finland. In total, 373 responses were obtained. In both cases, the participants were asked to discuss or respond to questions surrounding their health information behaviour, with a special focus on the Finnish National Health Archive Kanta and its portal My Kanta, yet not limited to it. Stojmenova et al. (2012) were concerned that older adults are not familiar with eHealth systems and hence cannot express their wants and needs. In this study, some experience of the My Kanta portal was a requirement for inclusion in the focus groups, whereas the survey participants may not have had such experience. On the other hand, occupational healthcare and municipalities in Finland generally have their own PAEHRs that were possibly familiar to most participants in the selected age group. Open-ended questions have been described as “any question where the respondent’s answers are not limited to a set of predefined response options” (Singer & Couper, 2017, p. 117). The following open-ended question was included in both studies: “If there would not be any limits in resources or available technologies when developing a digital health service, what would a best possible service look like and what functions would it have?”. In the focus groups, the question was asked at the end of the interview, whereas it in the survey was placed in the middle of the questionnaire, and altogether 160 respondents answered the open-ended question in the questionnaire. Focus group discussions were

recorded and transcribed verbatim, and for both interview transcripts and survey responses, the content analysis was inductive.

RESULTS

The responses provided to the open-ended question in the focus groups showed both similarities with and differences from the answers in the questionnaires. Overall, the suggestions in the focus groups were more in-depth and detailed than many of the written answers in the questionnaires, which were generally shorter than the spoken ones in the focus groups, and some were a mere "I don't know" or short expressions of being satisfied with the current services. However, most survey answers did contain practical suggestions.

As the focus group participants were given the open-ended question at the end of the interviews, they had already previously discussed several additions or improvements, and sometimes struggled to come up with additional new ideas, but the support from other group members helped to stimulate the envisioning, and some features that are not currently available were suggested. In one group, the participants discussed adding a health-promoting function, possibly by utilizing artificial intelligence to provide recommendations for eating or exercise to the user based on test results. In similar fashion, survey responses contained suggestions for personalized information, for example, in the form of recommendations for eating. Another focus group discussed reminders for check-ups at certain ages. Other suggestions in the group discussions were about possible improvements in access to the service, for example, in case one helps elderly parents who are not able to log in themselves, and also in the surveys there were suggestions for improved alternatives for authentication. Participants also requested channels for interactivity that were more varied than the current ones, including chats, and a wish that all information from different care providers could be gathered in one place, which were mentioned in both cases. Sometimes the used services did not fulfill information needs, as some participants wished for additional information on one's condition or a vocabulary of medical terminology in order to understand the language of the contents. There were also suggestions for a possibility to personalize the look of the service, for example, making it more pleasing to the eye or easier to read.

Ideas that occurred only in the focus groups were the possibility to remove obsolete information, for example, treatments in youth that are no longer of any use in older age, a section where the users themselves could add some possible important information, and a possibility to monitor one's own test results, for example through figures showing test values over time. In addition, information on vaccinations, blood type and allergies, to prevent possible trouble in case one is taken into care and is unable to inform about possible allergies, was mentioned.

In the surveys, some of the most often mentioned ideas included the possibility of contact via video connection and 24/7-access to health care. Furthermore, the possibility to make appointments and a well-functioning search function that would give access to needed information were mentioned. Other suggestions were the possibility to give feedback or comments on some medication, and some even envisioned a robot that could promptly perform a thorough examination and come up with a solution.

CONCLUSION

Our results can be useful for the design of eHealth services, or further development of existing PAEHRs, as they largely contain practical suggestions. The results of the open-ended question show that variations in information behavior are important to consider; older adults have needs for both more varied information and other communication channels than the PAEHRs they currently use can provide. Open-ended questions bring variation to the answers, as they are not necessarily tied to mere usability issues and testing of existing features, which is often the case in user-centered design, and not to features of only a certain service or application, either. Instead, respondents are allowed to innovate freely and propose ideas that go beyond specific services; hence, the answers can also be useful for different types of services. There are both advantages and disadvantages with this method. Envisioning an ideal service might be challenging, not the least in the middle of an otherwise mostly structured questionnaire, but at its best, it can also be a quite rewarding and interesting task. As questions are generally open-ended in interviews, including focus groups, answers to the intended topic might be given already in other parts of the interviews potentially diminishing the need for a specific question on design or development. On the other hand, the answers in our focus groups were more in-depth and suggested features in more detail than many of the answers in the survey, so collecting data through different methods might be the most useful solution. As it has been found that lack of resources hinder cooperation between end-users and software developers, and user-centered design approaches usually include several phases such as planning and testing requesting longer time spans and hence more resources, the option of including open-ended questions in larger studies related to eHealth information behavior, can be a cost-effective way to provide developers with needed feedback and make use of this form of citizen science that involves the intended users. This method has the benefit of not requesting a gathering of end-users at certain points of time at certain places, or the effort of testing existing services thoroughly. In order to provide software developers with these suggestions, researchers in the library and information science field should, furthermore,

strive to approach developers more directly to share their findings, instead of merely presenting or publishing them for an academic audience.

ACKNOWLEDGEMENTS

Available after review

REFERENCES

Comandé, G., Nocco, L. & Peigné, V. (2015). An empirical study of healthcare providers and patients' perceptions of electronic health records. *Computers in Biology and Medicine, 59*, 194-201.

Essén, A., Scandurra, I., Gerrits, R., Humphrey, G., Johansen, M.A., Kierkegaard, P., Koskinen, J., Liaw, S.-T., Ross, S.O.P. & Ancker, J.S. (2018). Patient access to electronic health records: differences across ten countries. *Health Policy and Technology*, 7(1), 44-56.

Gould, J.D. & Lewis, C. (1985). Designing for usability: key principles and what designers think. *Communications of the ACM, 28*(3), 300–311.

Hepworth, M. (2007). Knowledge of information behaviour and its relevance to the design of people-centred information products and services. *Journal of Documentation*, 63(1), 33-56.

Huvila, I., Enwald, H., Eriksson-Backa, K., Liu, Y.-H., & Hirvonen, N. (2022). Information behavior and practices research informing information systems design. *Journal of the Association for Information Science and Technology*, *73*(7), 1043–1057.

Kip, H., Keizer, J., da Silva, M.C., Beerlage-de Jong, N., Köhle, N. & Kelders, S.M. (2022). Methods for human-centered eHealth development: narrative scoping review. *Journal of Medical Internet Research* 24(1), e31858.

Luo, Y., Dozier, K., Ikenberg, C. (2021). Human-technology interaction factors associated with the use of electronic personal health records among younger and older adults: secondary data analysis. *Journal of Medical Internet Research, 23*(10), e27966.

Papoutsi, C., Reed, J.E., Marston, C., Lewis, R., Majeed, A. & Bell, D.(2015) Patient and public views about the security and privacy of electronic health records (EHRs) in the UK: results from a mixed methods study. BMC *Medical Informatics and Decision Making, 15*, 86.

Rytkönen, J., Kinnunen, U.-M. & Martikainen, S. (2022). Experiences of social and health care information system developers in co-operation with users. *Finnish Journal of EHealth and EWelfare*, *14*(2), 132–149.

Shah, S.G., Fitton, R., Hannan, A., Fisher, B., Young, T. & Barnett, J. (2015). Accessing personal medical records online: a means to what ends? *International Journal of Medical Informatics, 84*(2):111-118.

Singer, E. & Couper, M.P. (2017). Some methodological uses of responses to open questions and other verbatim comments in quantitative surveys. *Methods, Data, Analyses : a Journal for Quantitative Mehtods and Survey Methodology, 11*(2), 115-134.

Stojmenova, E., Imperl, B., Žohar, T., Dinevski, D. (2012) Adapted user-centered design: a strategy for the higher user acceptance of innovative e-Health services. *Future Internet*, *4*(3):776-787.

van Velsen, L., Ludden, G. & Grünloh, C. (2022). The limitations of user- and human-centered design in an eHealth context and how to move beyond them. *Journal of Medical Internet Research, 24*(10):e37341.

Vergouw, J.W., Smits-Pelser, H., Kars, M.C., van Houwelingen, T., van Os-Medendorp, H., Kort, H. & Bleijenberg, N. (2020). Needs, barriers and facilitators of older adults towards eHealth in general practice: a qualitative study. *Primary Health Care Research & Development, 21*, e54.

Wildenbos, G.A., Maasri, K., Jaspers, M., Peute, L. (2018). Older adults using a patient portal: registration and experiences, one year after implementation. *Digital Health, 4*, 1-9.

Wilson, T. D. (1981). On user studies and information needs. *Journal of Documentation, 37*(1), 3-15.